MoonWalker Weekly Meeting

10/30 - 11/04 Summary

Elijah, Rowel, Martin

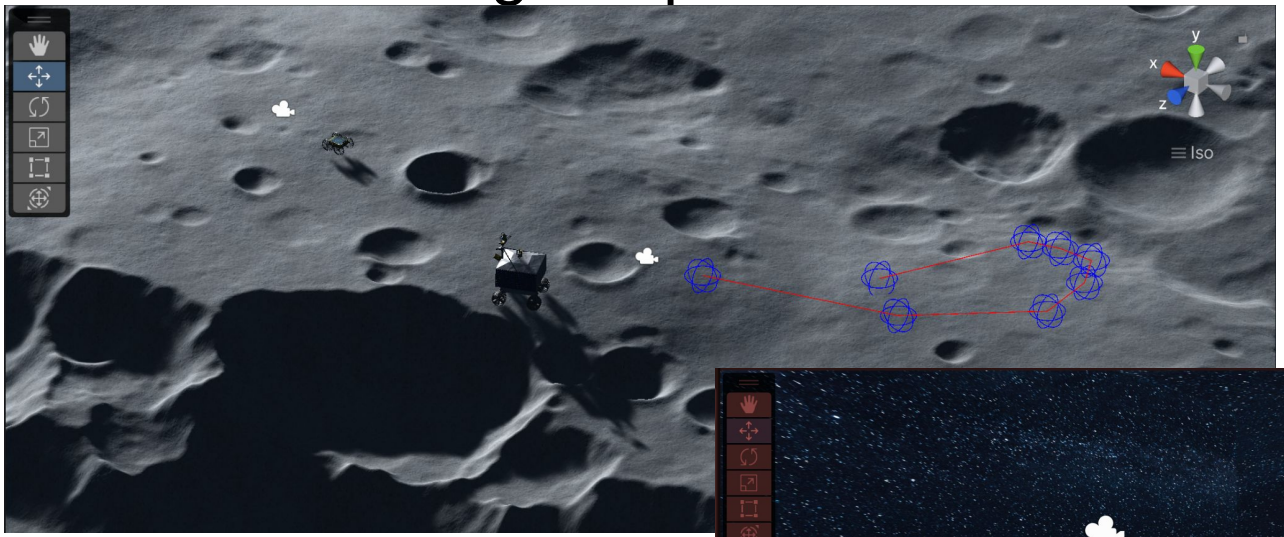
Progress Update

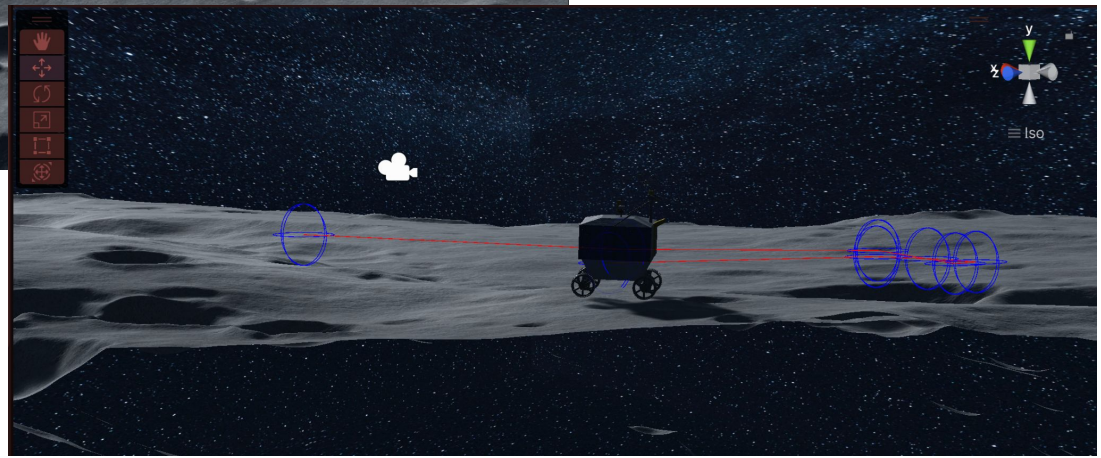
- Rowel
  - Fixed Digital Twin interoperability initial synchronization failure via Kafka
    - Possible Issue: Old session data (e.g stale, queued positional messages)
    - Solution: Reset the queued producer messages (Done manually)
      - Potential to automate but low priority as of now
  - Researched Channel Implementation
- Martin
  - Fixed interoperability coordinate/rotation issue for Rovers
  - Created and Improved Auto moving scripts
  - Automated Collision Testing
- Elijah
  - Fixed interoperability coordinate/rotation issue for Rovers
  - Finishing up BehaviorScript for Cadre
  - Automated Collision Testing

Old Auto-Moving Script: Problem

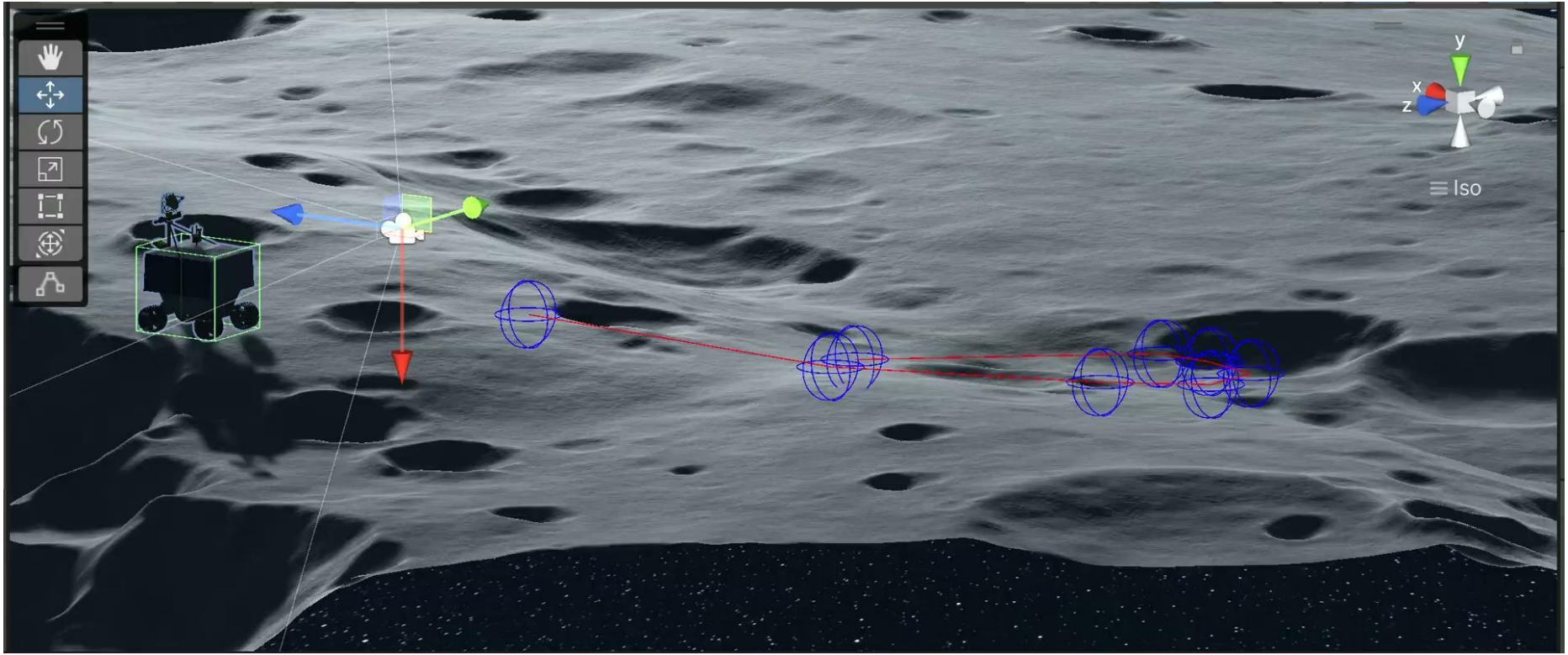
- Core Physics (Issue):
  - runs in Update()
    - Uses Update(), which just moves and turns the object each frame; Depends on frame rate
  - Moves/Rotates the transform directly with Vector3.MoveTowards and Quaternion.Slerp
    - No Rigidbody involvement; Doesn’t use Unity’s physics system (no Rigidbody).
    - Moves straight to the Waypoint’s full 3D position
  - Rotation Issue: Not shown in the video but the script originally flips the rover onto its side then moves
    - Fix: Apply a 90-degree rotation around the local Z-axis.

Old Auto-Moving Script: Unrealistic/Floats - Unity

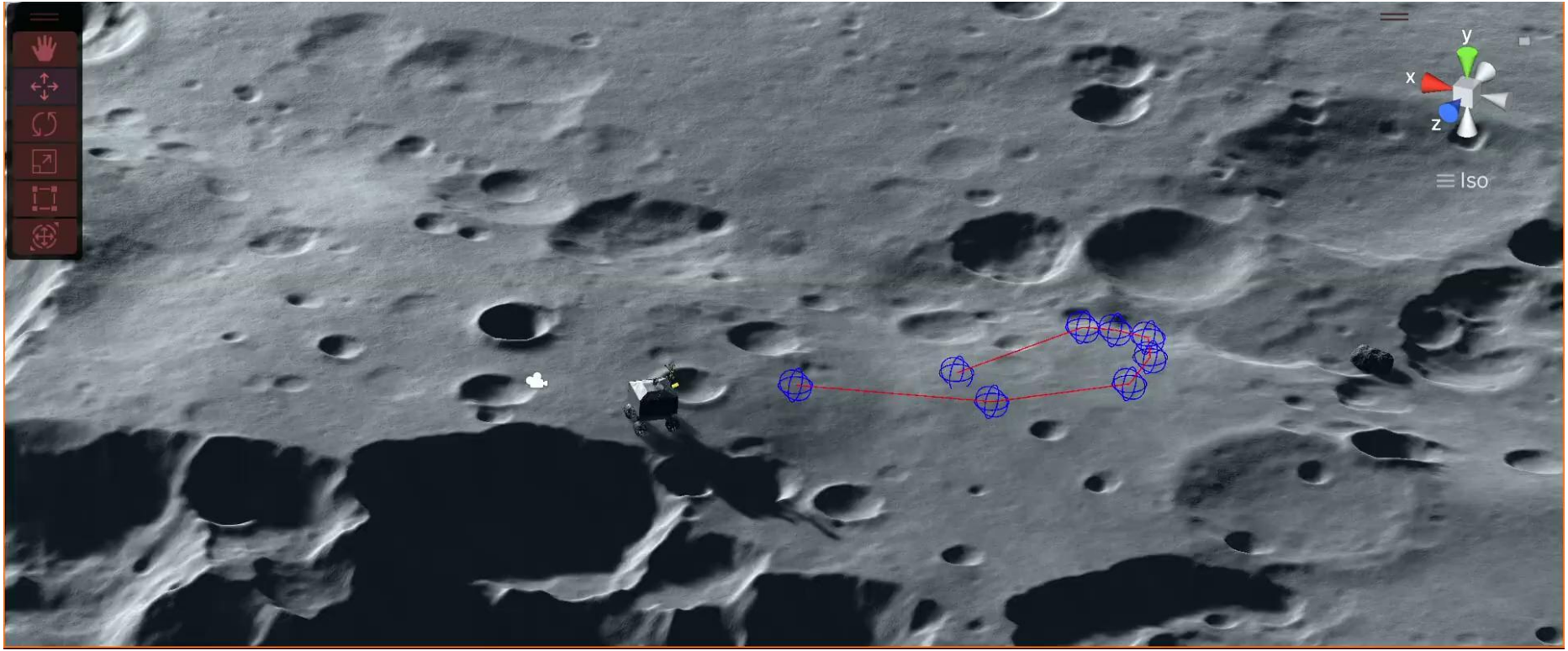




Old Auto-Moving Script - Unity



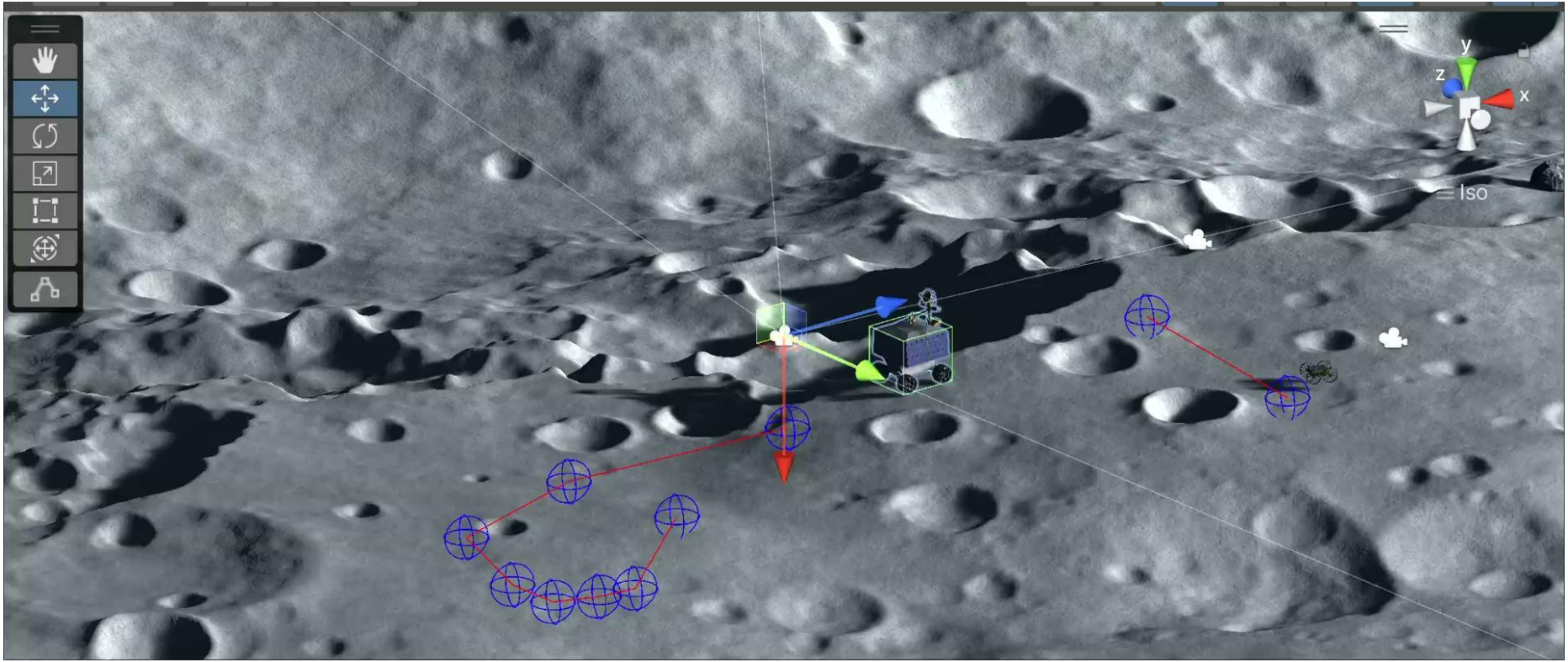
New Auto Moving Script: Improved Physics - Unity



New Auto-Moving Script: Solution

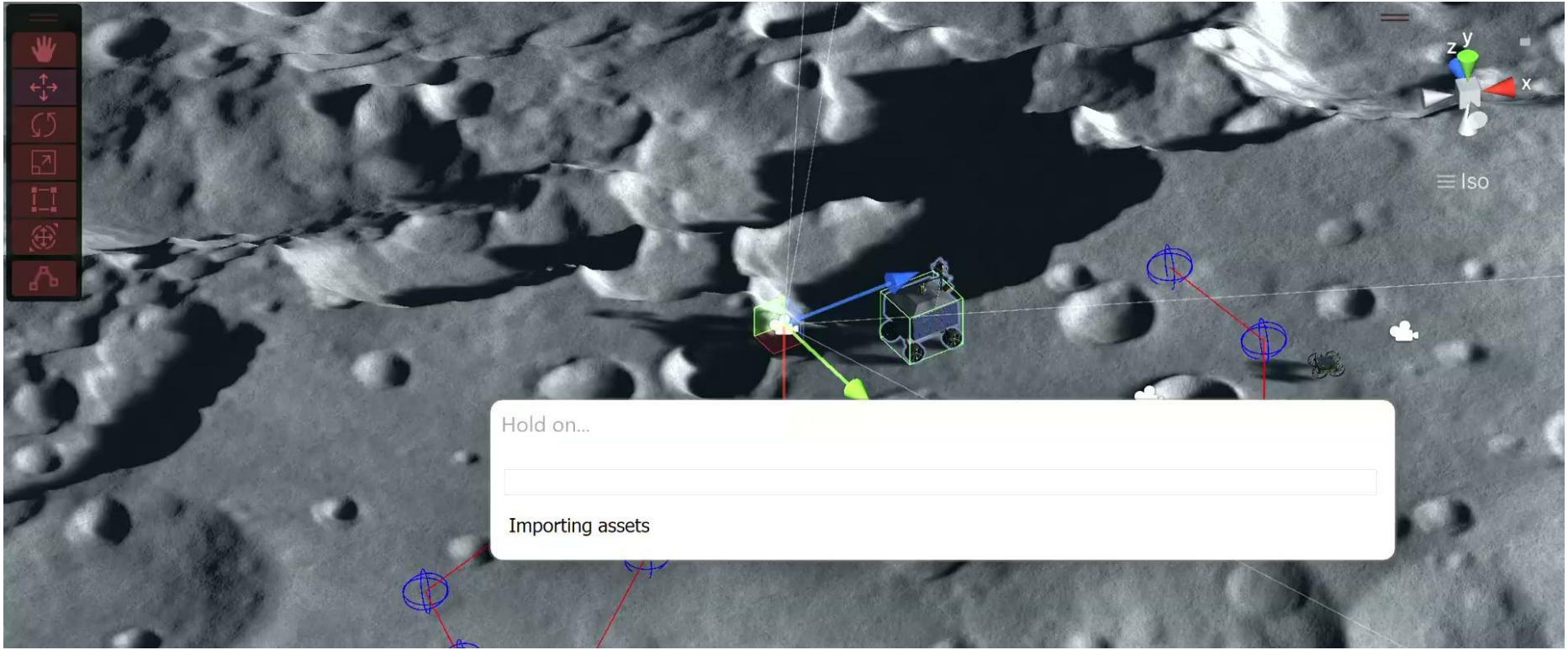
- Core Physics:
  - Runs in FixedUpdate()
    - At a fixed interval (default = 0.02 s = 50 times/sec)
  - Uses Rigidbody.MovePosition / Rigidbody.MoveRotation; Uses physics
  - Intentionally ignores the waypoint's Y and follows the terrain slope
    - Raycasts down for a ground normal, then projects the "to waypoint" vector onto that plane
  - Computes "virtual inputs": Replicate manual driving script physics
    - Throttle that eases near the waypoint
    - Turn input used for wheel visuals

Collision Testing I w/ Improved Script - Unity

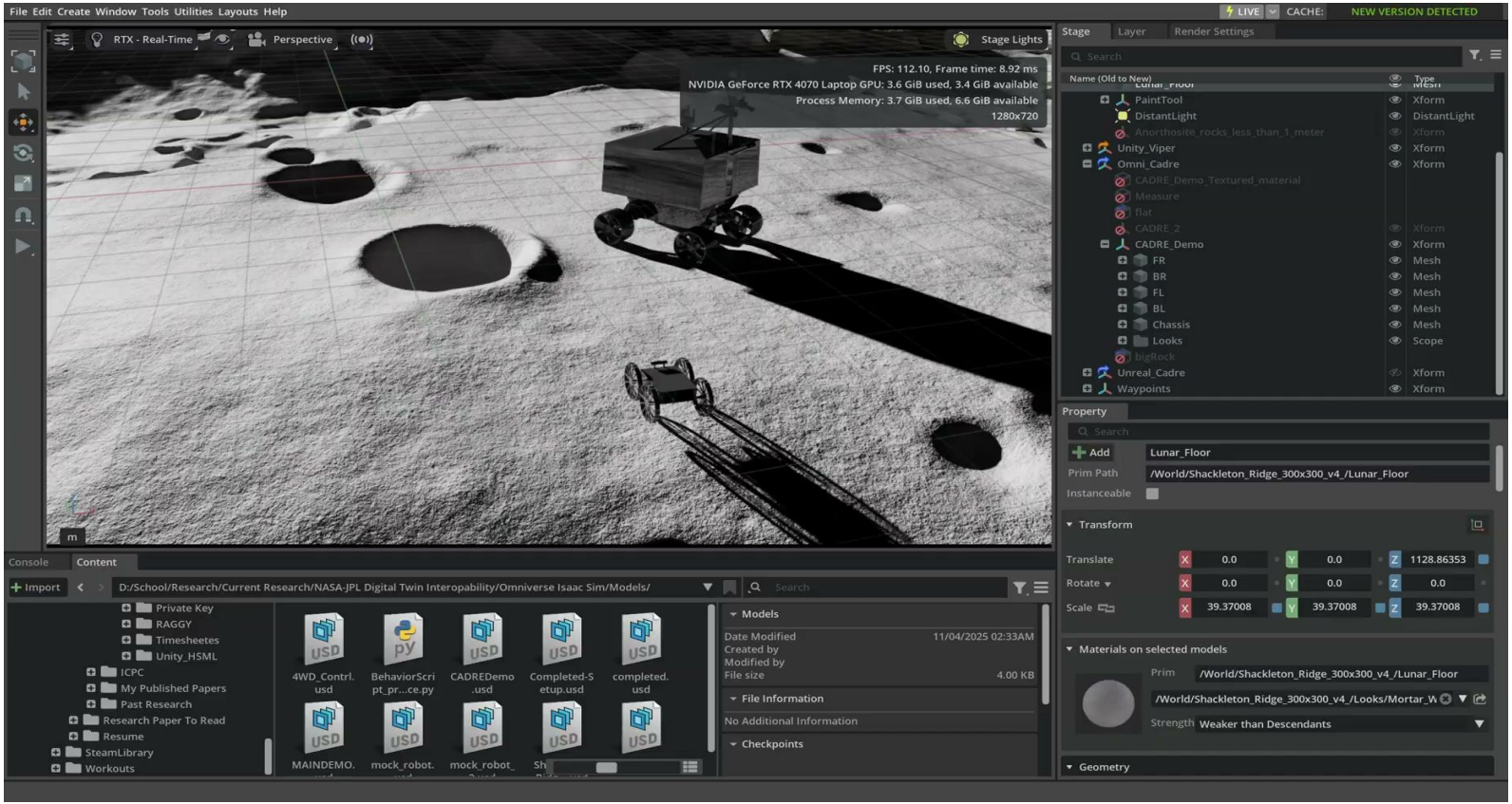


Collision Testing I w/ Improved Script - Omniverse

Collision Testing II w/ Improved Script - Unity



Collision Testing II w/ Improved Script - Omniverse



Action Plan

- Rowel
  - Continue working on channel feature
    - Update Database for Channel Implementation
    - Update code
      - API Registration, app.py, HSML Schema
- Martin
  - Create more collision tests for reproducibility purposes at different locations
  - Work w/ Elijah and Rowel on replicating collision performance in each simulator
    - Low priority: Improve the auto-moving script: slightly janky interaction with Terrain
    - Low priority: Rover falls through some areas of map; Identify areas and try to fix.
    - Documenting local configurations
- Elijah
  - Create more collision tests for reproducibility purposes at different locations
    - Finish up BehaviorScript
  - Work w/ Martin and Rowel on replicating collision performance in each simulator
  - Documenting local configurations